

#VLSM

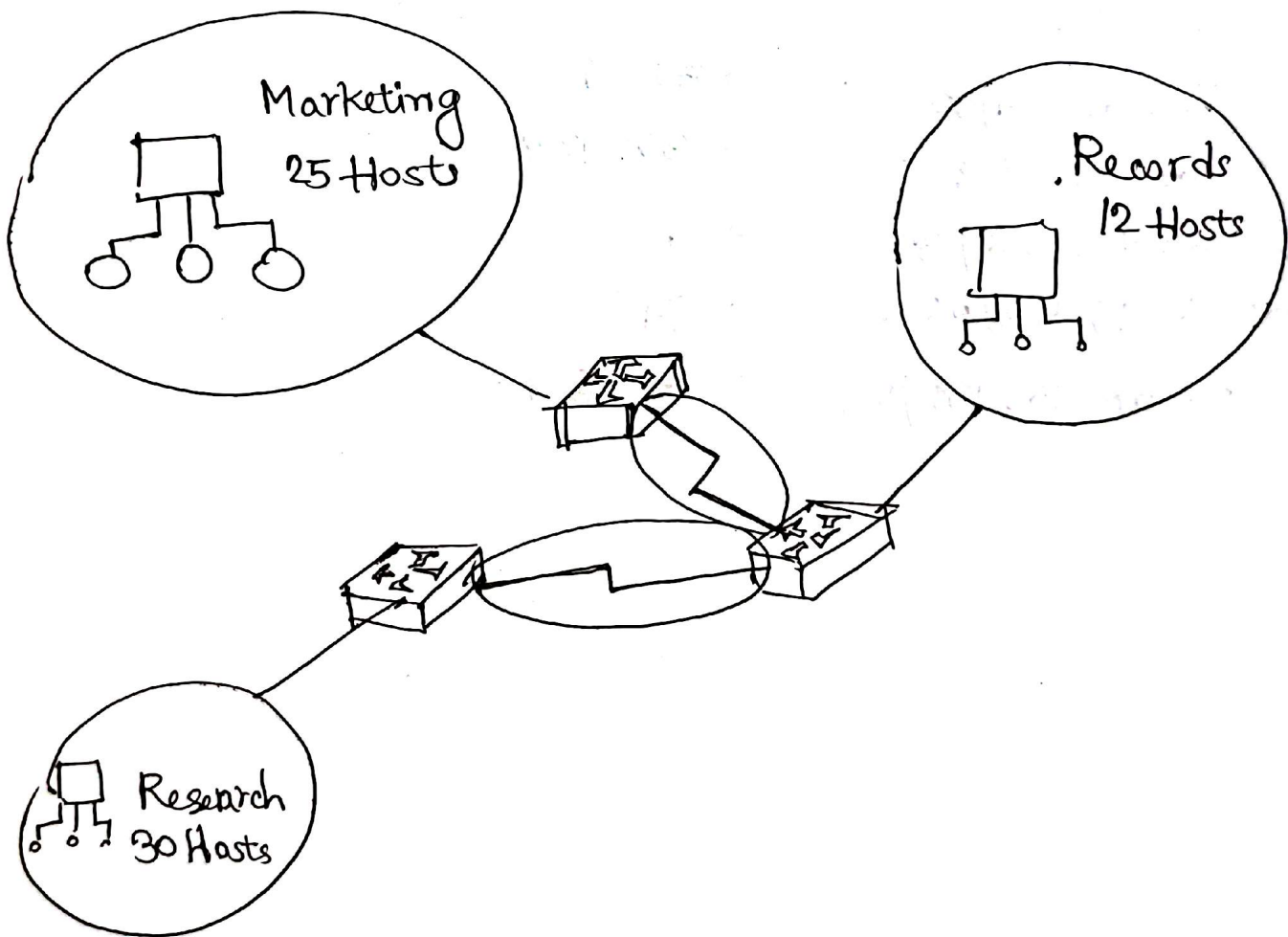
'Variable Length Subnet Mask'

#FLSM

'Fixed Length Subnet Mask'

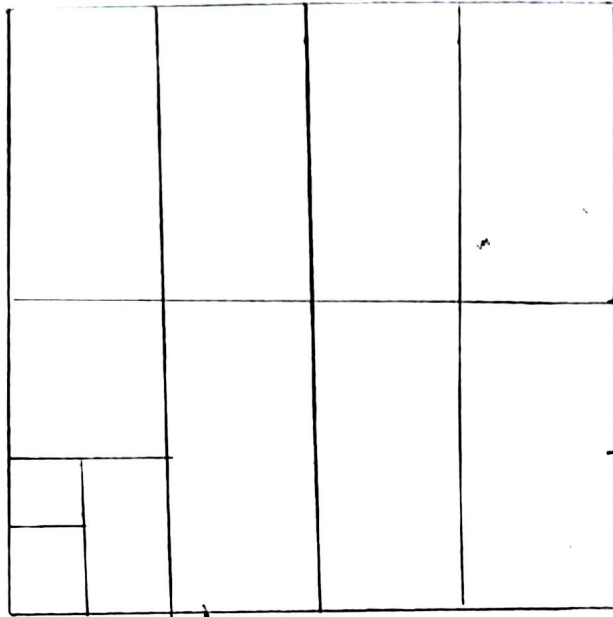
Classful Subnetting Example:

IP Address : 192.168.1.0 /24



Shayon

32, 32, 16, 4, 1



$$32 - 27 = 5$$
$$2^5$$

Subnets : 5

$$192.168.16.224/30 \longrightarrow \text{4}$$

$$192.168.16.0/25 \longrightarrow \text{32} \quad 128$$

$$192.168.16.228/30 \longrightarrow \text{4}$$

$$192.168.16.128/26 \longrightarrow \text{64}$$

$$192.168.16.192/27 \longrightarrow \text{64}$$

#Routing Algorithm Classification ^{Shaparo}



global: all routers have complete topology, link cost info

link state: algorithms

How fast
do routes
change

static: routes
change
slowly over
time.

dynamic:
routes change
more quickly

- periodic updates or in response to link cost changes.

decentralized: iterative process of computation, exchange of info with neighbours.

→ routers initially only know link costs to attached neighbours

→ distance vector algorithms

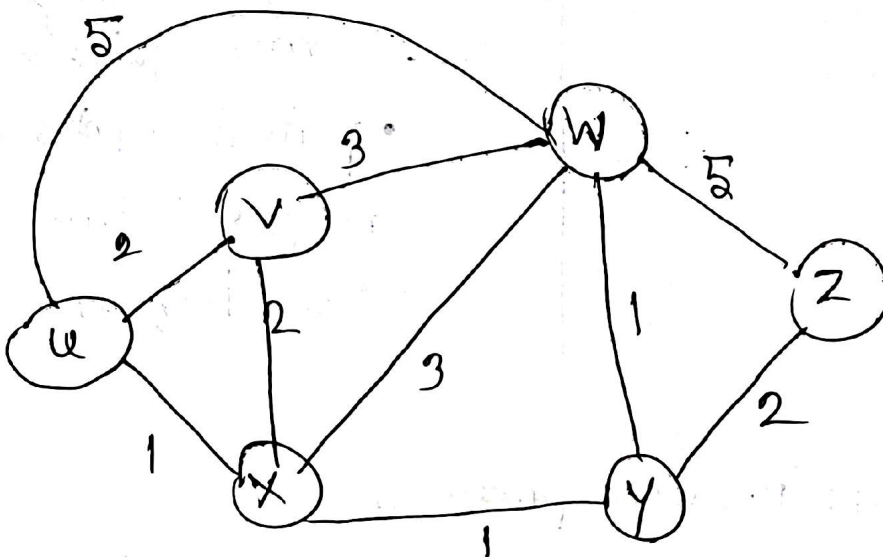


Dijkstra's link-state routing algorithm

$N' = \{u\}$
for all nodes v
...

Dijkstra's algorithm example

 PDF / Slides

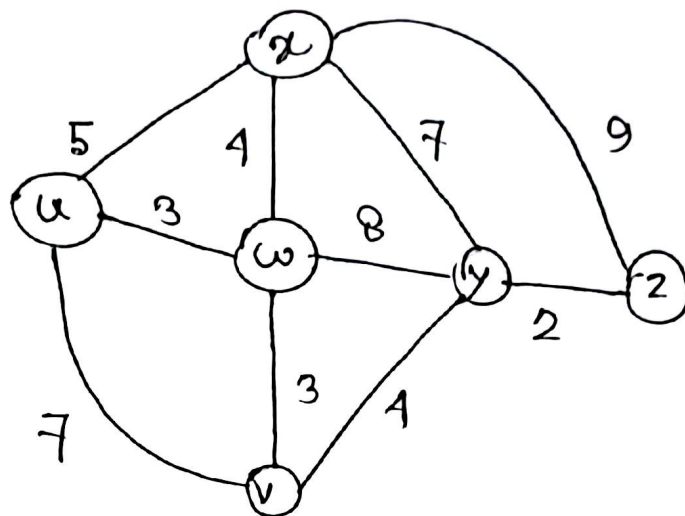


find a not in N' such that $D(a)$ is a minimum
add a to N'

update $D(b)$ for all b adjacent to a and not
in N'

$$D(b) = \min(D(b), D(a) + c_{a,b})$$

Dijkstra's Algorithm: Another Example



Step	N'	v $D(v),$ $P(v)$	w $D(w),$ $P(w)$	x $D(x),$ $P(x)$	y $D(y),$ $P(y)$	z $D(z),$ $P(z)$
0	u					
1						
2						
3						
4						
5						

Dijkstra's Algorithm

Initialization

$$N' = \{u\}$$

for all nodes v

if v adjacent to u

$$\text{then } D(v) = c_{u,v}$$

$$\text{else } D(v) = \infty$$

→ loop

find w not in N' such that $D(w)$ is a minimum

add w to N'

update $D(v)$ for all v adjacent to w and not in N' :

$$D(v) = \min(D(v), D(w) + c_{w,v})$$

loop until all nodes in N

complexity : n nodes

$n(n+1)/2$ comparisons $O(n^2)$ complexity

more efficient implementations possible : $O(n \log n)$

each router must broadcast its link state information to others n routers.

efficient (& interesting) broadcasting algorithms: $O(n)$ link crossings to disseminate a broadcast message from one source.

when link costs depend on traffic volume, route oscillations possible.

Sample Scenario:

- A. Routing to Destination A,
- B. Traffic Entering at D, C, E
With rates 1, $e(<1)$, 1
- C. Link costs are directional & volume dependent

Distance Vector Algorithm

Based on Bellman Ford (Dynamic Programming)

Let $D_x(y)$: cost of least-cost path from x to y :

Then:

$$D_x(y) = \min_v \{ C_{x,v} + D_v(y) \}$$

min taken over all neighbors v of x

??

??

Bellman Ford Example

→ PDF